

Valcore System Architecture

Client layer

Web Browser
upload CSV · view dashboard

HTTP POST
/predict

rendered
report

Application layer (Flask · app.py)

Upload
Validation

Preprocess
& Encode

Random Forest
Inference

Threat
Assessment

Template
Rendering

Model artifacts (loaded at startup)

model.pkl

encoders.pkl

feature_columns.pkl

Offline
training
produces
artifacts

Presentation

templates/index.html

static/script.js (Chart.js)

static/style.css

